

Introduction

EC 325: Econometrics

Prof. Caraher

Colby College

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About Me: Prof. Ray Caraher

I am a recent PhD from UMass Amherst. My research focuses on the relationship between public health and maternal, infant, and child health and health disparities.

This is my first year at Colby!

- Email: rcaraher@colby.edu. I will do my best to respond within 24 hours on weekdays.
- Office: 365 Diamond Bldg.
- Office Hours: Tuesdays, 2pm - 4pm, Thursdays, 2pm - 3pm, Fridays, 10:30am - 12:30pm. Or by appointment.

What is Economet

What is this Class About?

Econometrics: “...statistical methods for estimating economic relationships, testing economic hypotheses, evaluating and implementing government policy.”

What is this Class About?

Or, in my opinion,

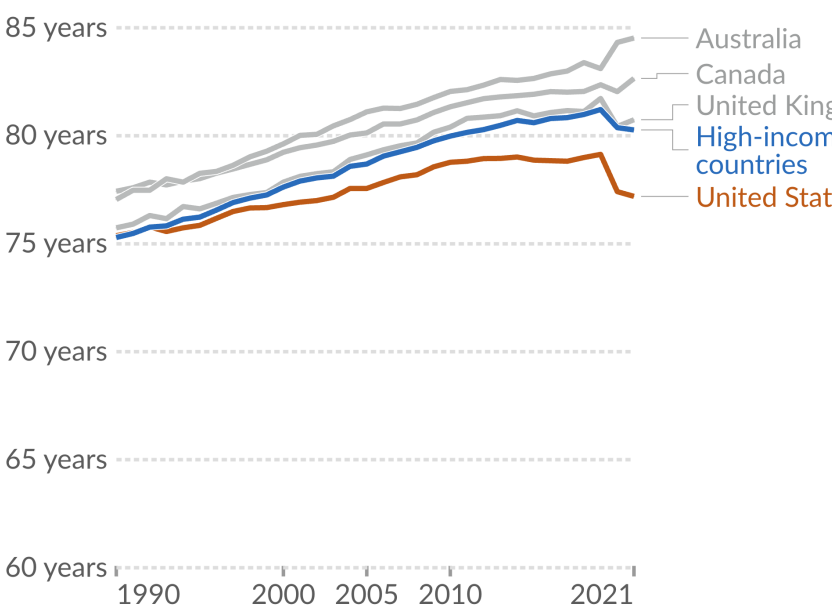
Econometrics: The practice of using statistical methods to study social phenomena and put it into public policy

Modern econometrics has really focused on **causal effects** of social phenomena/policy
correlations

The US is Unhealthy Relative Countries

Life expectancy in the United States is lower than peer nations

The period life expectancy at birth, in a given year.

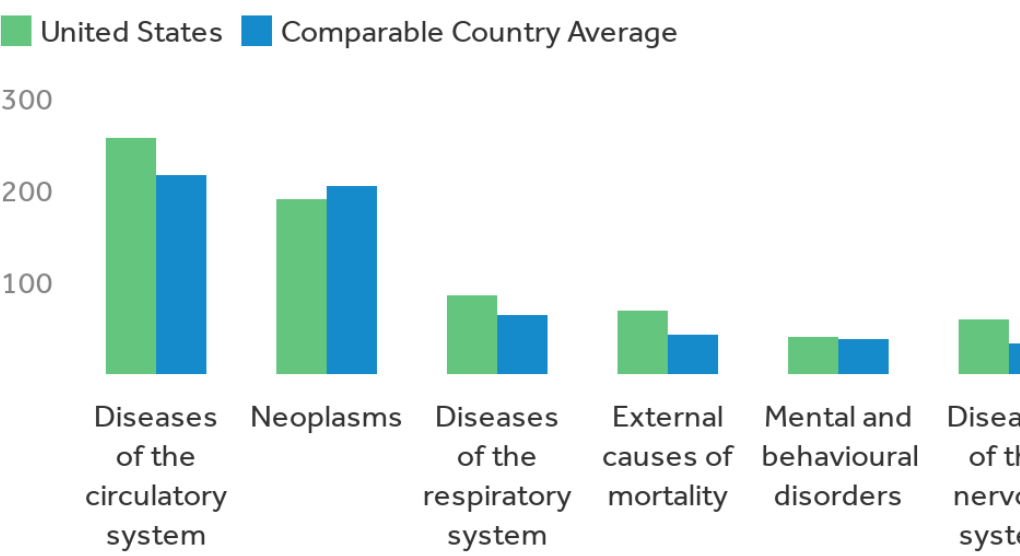


Data source: UN WPP (2022); HMD (2023); Zijdeman et al. (2015); Riley (2005)
OurWorldInData.org/life-expectancy | CC BY

Life expectancy in the US vs. peer countries

The US is Unhealthy Relative to Other Countries

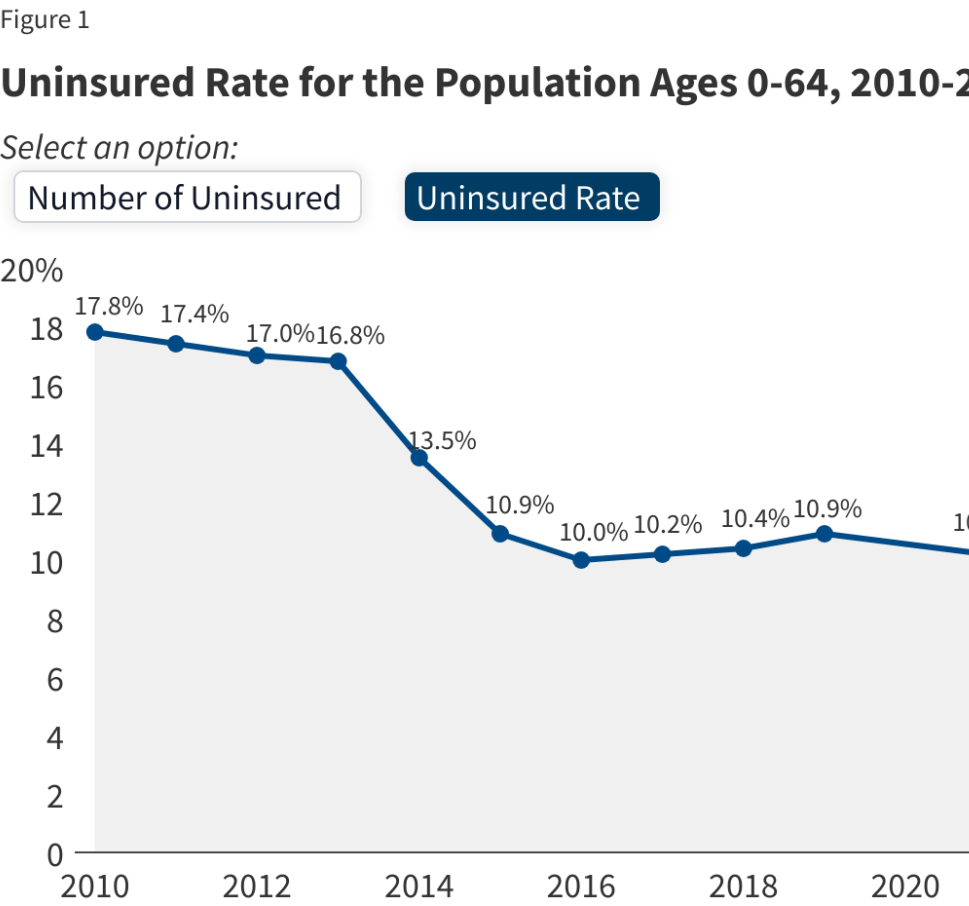
Age-adjusted major causes of mortality per 100,000 population



Note: Data for Canada are from 2013 and France are from 2014

Source: KFF Analysis of OECD Health Statistics (Database)

One potential reason: health



Note: Due to disruptions in data collection during the first year of the pandemic, the Census Bureau did not release ACS 1-year estimates in 2020. Includes individuals ages 0 to 64.
Source: KFF analysis of 2010-2023 American Community Survey, 1-Year Estimates

Uninsured rates overtime

Does having health insurance to be healthier?

Discuss with a classmate

Does having health insurance make you healthier?

- Health insurance coverage is **correlated** with better health outcomes
- But does that mean health insurance **causes** better health?

Does having health insurance make you healthier?

- Health insurance coverage is **correlated** with better health outcomes
- But does that mean health insurance **causes** better health?

Econometrics can help us answer this question

Other examples of causal inference questions

- Do tariffs increase domestic production?
- Does a college degree increase your earnings?
- Is immigration bad for native workers?
- Do police reduce crime?

What Is the Broad Process for Econometric Analysis?

Let's say we want to know what the effect of training has on your wage.

What **steps** would we need to take to answer this question?

What Is the Broad Process for Econometric Analysis?

Step 1: Write an *econometric model*

- Sometimes formally derived from an *academic model* (e.g., profit maximization of the firm)
- Always based on some **intuition** which **relates** some *variable(s)* to some other variable

What Is the Broad Process for Econometric Analysis?

One example of an econometric model we will learn what this equation means later!):

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper +$$

What Is the Broad Process for Econometric Analysis?

Step 2: Find and explore the data (easier said than done)

Many different types of data:

- cross-sectional (most common): a sample of individuals, households, firms, countries, or any other entity measured at a given point in time
- Data for econometric is organized (or not organized) as a **matrix** or **dataframe**

What Is the Broad Process for Econometric Analysis?

```
# A tibble: 4 × 5
  person wage educ exper training
  <dbl> <dbl> <dbl> <dbl>    <dbl>
1     1  17.2   12    10         2
2     2  11.9   16     1         0
3     3  23.2   16     2         1
4     4  18.4   10    15         3
```

What Is the Broad Process for Econometric Analysis?

- Are we missing a lot of values for our data?
- Are there outliers, obvious data-entry errors, or other oddities we need to deal with?
- What are some summary statistics (mean, standard deviation, etc.) for our variables?
- How do certain variables of interest move together (or not)?

What Is the Broad Process for Econometric Analysis?

Step 3: *Estimate* the relationship

- Here we use statistical techniques to **estimate an econometric model** using our data
- Many ways to do this, but by far most common is **Least Squares (OLS)**
- Also estimate the **uncertainty** of our estimates and test for *statistical significance*

What Is the Broad Process for Econometric Analysis?

```
Call:
lm(formula = wage ~ educ, data = test_data_2)

Residuals:
    Min       1Q   Median       3Q      Max
-0.42418 -0.04713 -0.00786  0.04971  0.29730

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.199694    0.070398   17.04  <2e-16 ***
educ         0.118062    0.006873   17.18  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1116 on 40 degrees of freedom
```

What Is the Broad Process for Econometric Analysis?

Step 4: *Interpret* the estimates

What does our estimate mean?

- There is a literal, *mathematical interpretation*
- In our made-up estimate above, each additional year of education suggests an increase in wages
- But what does this really mean? Is it big? Is it economically important?

What Is the Broad Process for Econometric Analysis?

Step 4: *Interpret* the estimates

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In this course, we will learn how to do **all** of this

Econometrics with Statistical Software

Statistical Software

Much of the econometric process is done using specialized software

We use this software to:

1. “Clean” our data and organize it so we can do econometric analysis
2. Use our data to explore our questions using descriptive statistics and visualizations
3. Estimate our models and calculate the statistical uncertainty
4. Output our results as tables and figures so we can write/present

Examples of Statistical Software

- **Stata:** Proprietary software purpose-built for econometrics
- **R:** Open-source source programming language with a focus on statistical analysis
- **Python:** Open-source source programming language with a focus on data-science more broadly
- **EViews:** Proprietary software with a focus on econometrics

R and R Studio

In this class, we will focus our learning on the **R** programming

Why R?

- **Open-source:** Free to use, with a large community which thousands of packages to extend the capabilities of R
- **Object-Orientated:** Can store multiple “objects” in its memory, providing flexibility in cleaning, organizing, and compiling data
- **Not Just Econometrics:** Specialized uses of R in dozens of industries, including the private sector

R Studio: An integrated development environment (IDE) designed for the R programming language. It provides a user-friendly interface and tools that make it easier to write, run, and manage R code and projects.

R and R Studio

In this class, we will focus our learning on programming language

R and R Studio

There are instructions on how to install the appendix of the Lecture Notes. Review the and R Studio as soon as you can!

Learning Econome

EC 325

What We Will Learn in This Course

- The “Gold Standard” of Causality: Randomized Control Trials
- Econometric Swiss Army Knife: Ordinary Least Squares (OLS)
- Controlling for Confounders: Multivariate OLS
- How Uncertain is Our Estimate? Statistical Inference Tools
- The Non-Linear World: Dummies, Interactions, and Polynomials
- Panel Data and Fixed Effects: On the Road to Causal Inference
- The Causal VIP: Difference-in-Differences
- Alternative ways of causal methods: instrumental variables
- Time Series Methods

TA Info

TA:

- Amrit Shakya (ashaky26@colby.edu)

TA Help Hours:

- TBD

Required Materials

- Reliable access to a computer (talk to me if this is a concern!)
- A [Github](#) account to access the course materials
 - A free account is fine, but you get access to more features at the Pro level for free if you use your @colby email. If you do, send them your ID!

Git and GitHub

I use **Git** and **GitHub** to host all course content (including assignments, which are on Moodle)

- Includes lecture notes, slides, data, in-class exercises, etc.
- **Git**: A version control system that tracks changes to files over time
- **GitHub**: A website that hosts Git repositories, making them easy to share

Git and GitHub

Why use Git/GitHub?

- Industry standard for collaborative code control
- You can see exactly what changed between versions
- Useful skill for research and industry jobs

Lecture Notes

- You are required to read the lecture notes *before we cover the unit in class*
- The notes are available either in the Git repository or on the website <https://www.raymondcaraher.com/ec325/>
- Lecture notes include optional multiple choice questions

Optional Texts

- These textbooks are easy to find and are this course content is
- **Introductory Econometrics: A Modern Approach by Jeffrey Wooldridge**
The 7th edition is the most recent, but any edition 5 or above is fine. We will primarily use this text for part of the course more reliant on the theory.
- **Real Econometrics by Michael Bailey (2020)**. This is a good alternative to the Wooldridge textbook.
- **Mastering 'Metrics by Joshua D. Angrist and Jörn-Steffen Pischke**
is an excellent companion to the other texts, but is not like the others.

Pre-Reqs

- Intermediate Micro
- Research Methods and Statistics

If you don't have either one of these, let me

Assignments

Your grade will be determined by:

Assessment	Weight
Concept Checks	10%
Problem Sets	30%
Midterm Exam	10%
Final Exam	15%
Final Project	30%
Engaged Learning	5%

Concept Checks

- Short, weekly coding assignments testing understanding of the readings and lectures
- Due each Friday at 11:59pm (with some exceptions)
- Graded on a 3-point scale (0 = no attempt)
- Lowest 2 grades dropped at end of semester

No collaboration allowed on concept checks
hours/email if you have questions!

Problem Sets

- 4 problem sets during the semester
- Designed to have you **replicate results** from **published economics papers**
- A considerable portion of your grade — mini-project
- No late problem sets allowed

You can (*and are encouraged!*) to work in groups on problem sets.

AI Policy

AI tools can be useful for coding questions, but they can become a crutch. The AI policy for this course is as follows:

- You are allowed to use AI tools on your **final project for specific, coding-related questions**
- **No AI tools allowed on Concept Checks**
- Any use of an AI tool must be **attributed** in your submission
- You **must understand** the code you submit

If you choose to use AI, Colby recommends you log in **with your Colby account**.

Exams

- Midterm exam (in-class, Thursday, March 14)
- Final exam (per registrar schedule)
- Both are in-person; one double-sided notes
- Focus on concepts and interpretation – you will analyze output

Engaged Learning

Active learning is **necessary** to learn econ counts as part of your grade.

What counts:

- Coming to class and answering/asking questions
- Going to office hours and TA help hours
- Coordinating among group members
- Helping classmates with the course material

Attendance

- All classes are **in-person** (unless otherwise due to weather) and attendance is **required**
- Please **do not come to class** if you are feeling ill
- No formal doctor's note required, but please check-in
- Unexcused absences will count against learning grade

Final Project

A research project completed in **groups** or
will form groups based on your shared sub
then groups will narrow down on a specific
question.

This project will be a **considerable amount**
have check-ins throughout the semester to
are going smoothly.

Final Project (cont.)

You will have to turn in:

1. A project proposal (week 5)
2. A data summary (week 8)
3. A preliminary findings report (week 10)
4. A final presentation (last week of class)
5. A final written report (last day of finals)
6. A peer review evaluating your and your contributions (last day of finals)

About you!

1. Name
2. Pronoun
3. Year
4. What you did over Jan Plan

For Next Time

- Concept Check 1 (due tomorrow!)
- Fill out the survey so I can assign groups
- Read **unit 1** and **unit 2** of the lecture notes